

m2sv: A Scalable Benchmark for Map-to-Street-View Spatial Reasoning

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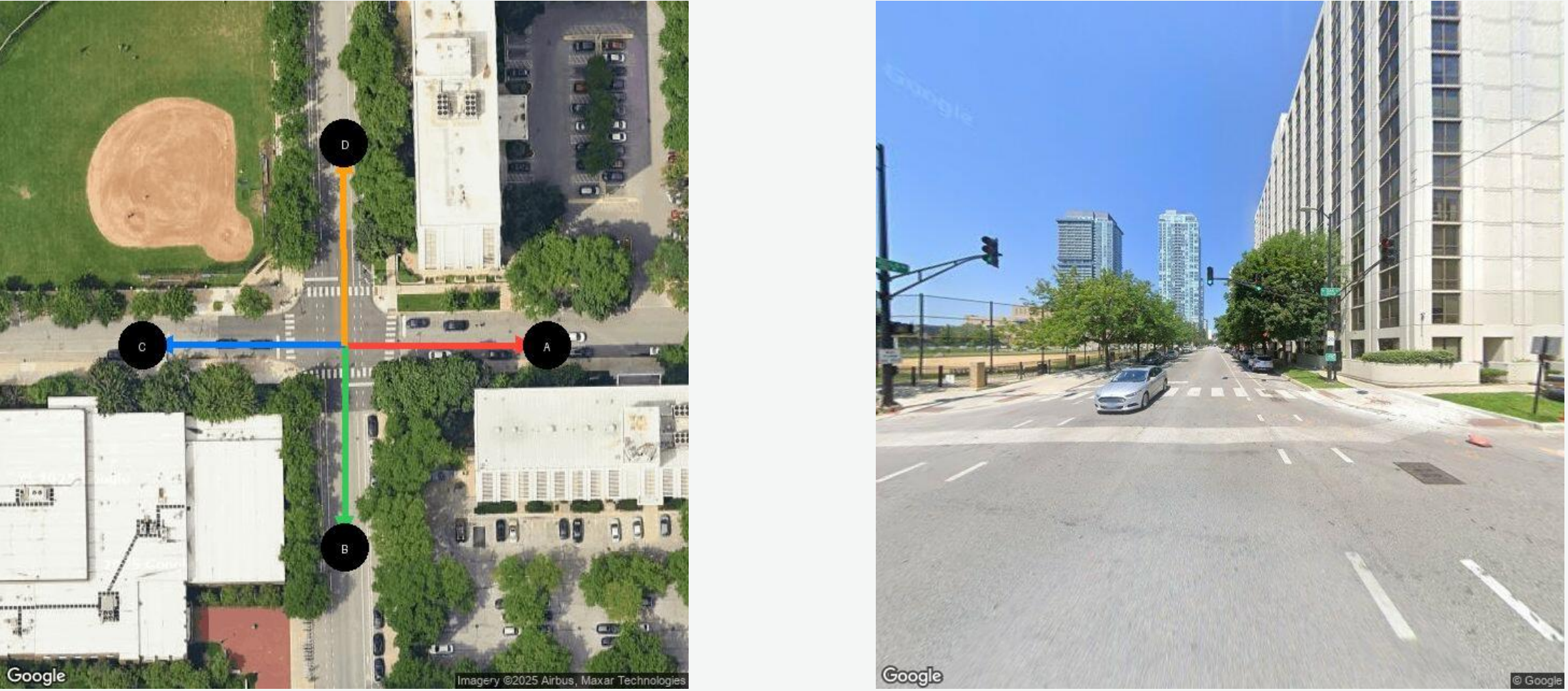
CTB Workshop @ ICML 2026 · **non-archival**

github.com/YosubShin/m2sv

live demo: m2sv.yosubshin.com

The Task

Given a **north-up overhead map** with labeled candidate directions and a **Street View photo** taken at the same intersection, infer *which labeled direction the camera was facing*. The camera sits at the intersection center; letters point outward along each road.



Solve by aligning stable map geometry (roads, parcels, landmarks) with the ground-level view — not transient cues.

Benchmark & Data (open)

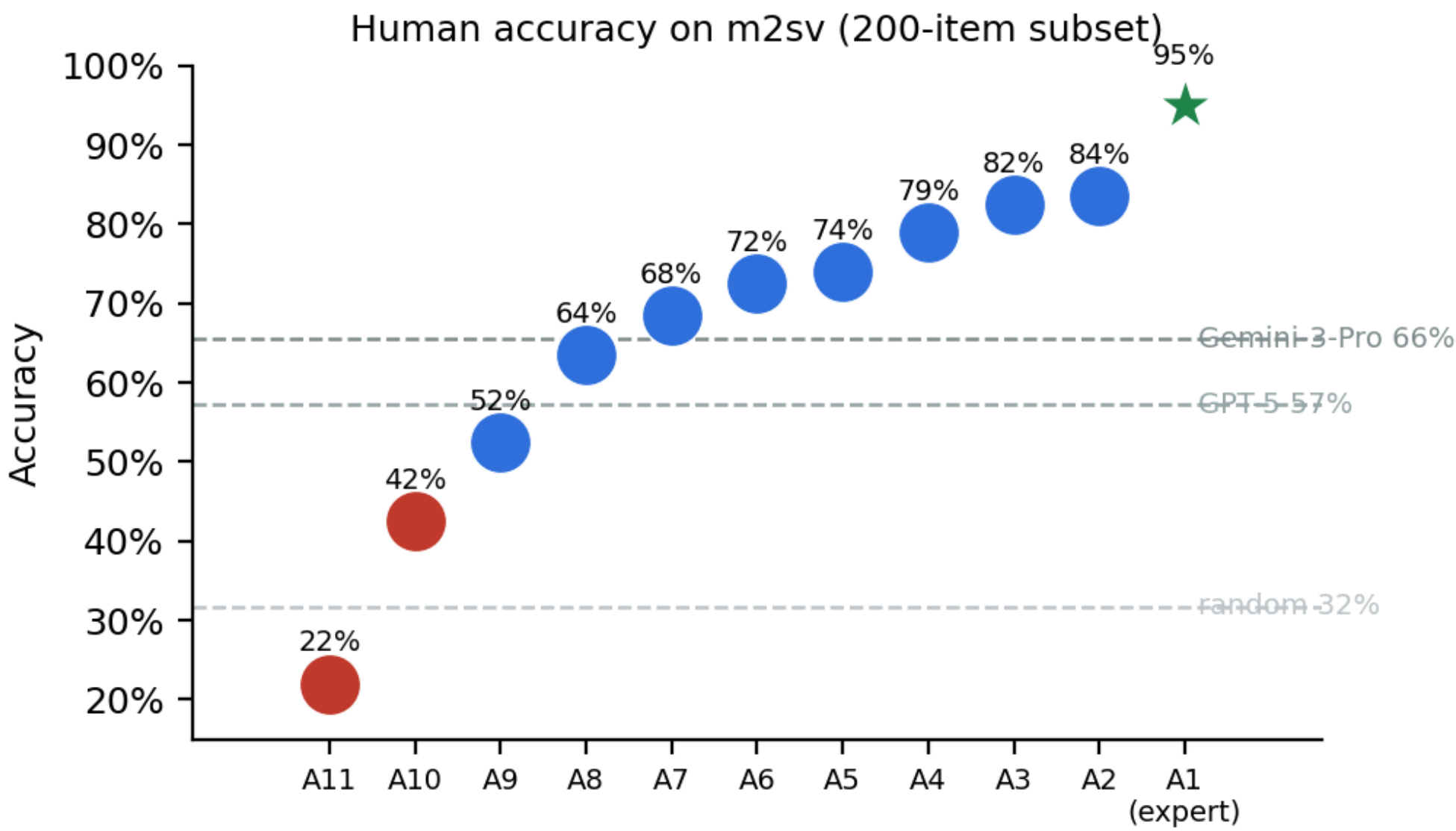
- **m2sv-20k** — 20,000 real intersections (10k train / 10k validation) across **32 cities**, multiple continents, controlled candidate ambiguity.
- **m2sv-sft-11k** — 11k curated structured reasoning traces for SFT.
- Released as **blueprints** (coordinates + azimuths) plus rendering code to respect map licensing; all eval/training code public.

Zero-shot: humans clear the best VLM

Model	N	Acc. (%)
Gemini-3-Pro	1k	65.2 ± 3.0
GPT-5	1k	57.2 ± 3.1
Gemini-2.5-Pro	1k	47.2 ± 3.1
Qwen3-VL-235B (Think)	1k	42.7 ± 3.1
Random baseline	1k	31.4 ± 2.9
Human (engaged, n=8)	200	72.0 ± 9.7
Human (expert)	200	95.0

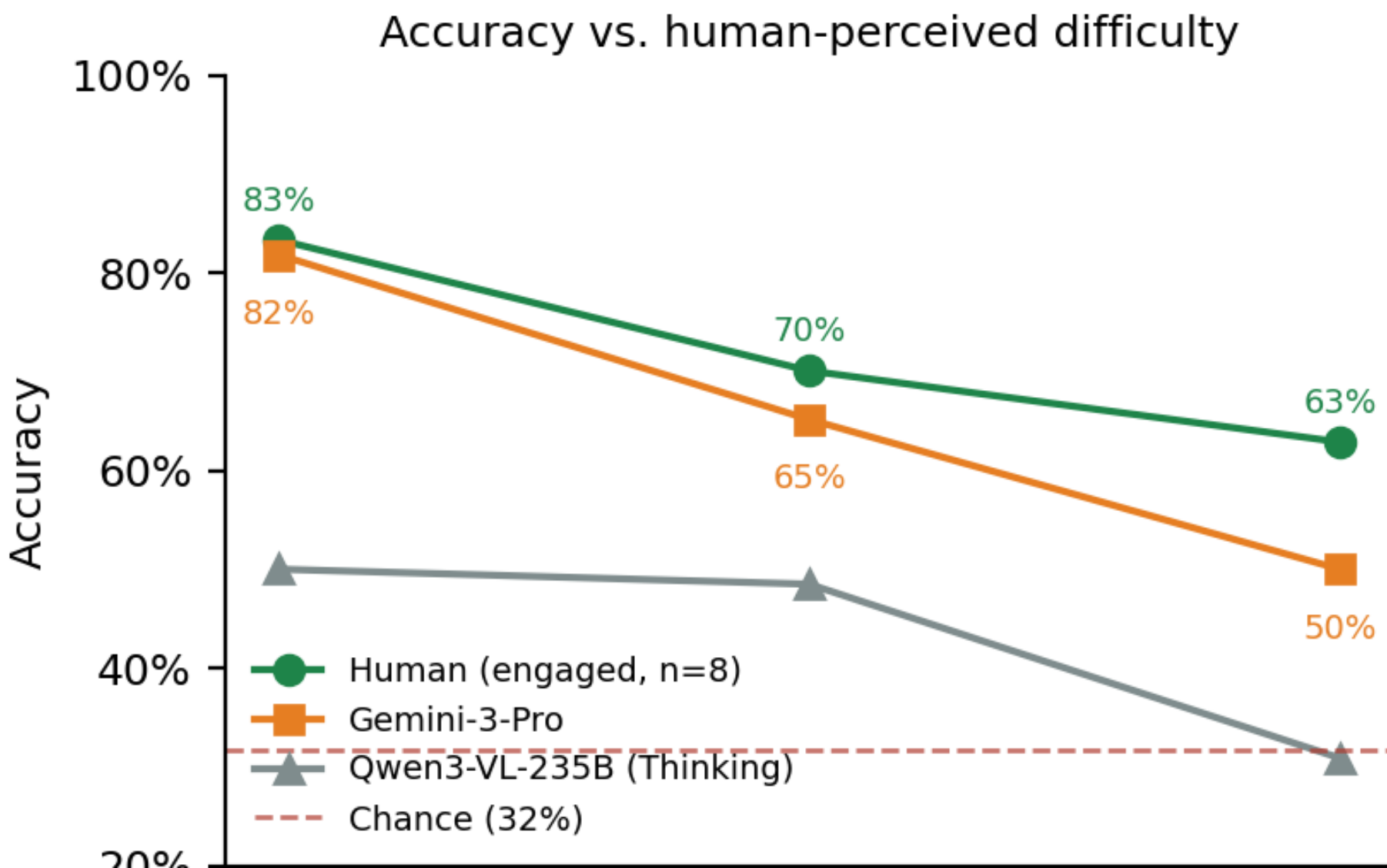
Multi-annotator study (12 participated; the 10 who finished all 200 are analyzed). Two low-agreement outliers excluded by *agreement* ($\kappa_{\text{expert}} < 0.3$), not accuracy.

Human baseline & agreement



Attentive humans converge (median pairwise $\kappa = 0.48$, up to 0.76); Gemini-3-Pro sits *below* the human consensus (median $\kappa = 0.42$), Qwen3-VL-235B lower still (0.15). The gap shows up in *agreement structure*, not only accuracy.

Difficulty: the hardest-tier collapse



Why models fail (qualitative analysis)

Humans solve nearly all of these. Model errors come from *brittle alignment strategies and reliance on unstable evidence*, not missing information. Each card: overhead map, street view, a trace excerpt, and the error.

Ego↔allo inversion

GT: A Pred: C



“Looking from the center towards A, the road curves to the right — contradicting the left curve in the photo.”

Why: The map shows a *left* curve, but the model reasons as if it were a right curve. Left/right inversion at the map↔photo boundary.

Over-reliance on unstable cues

GT: B Pred: D

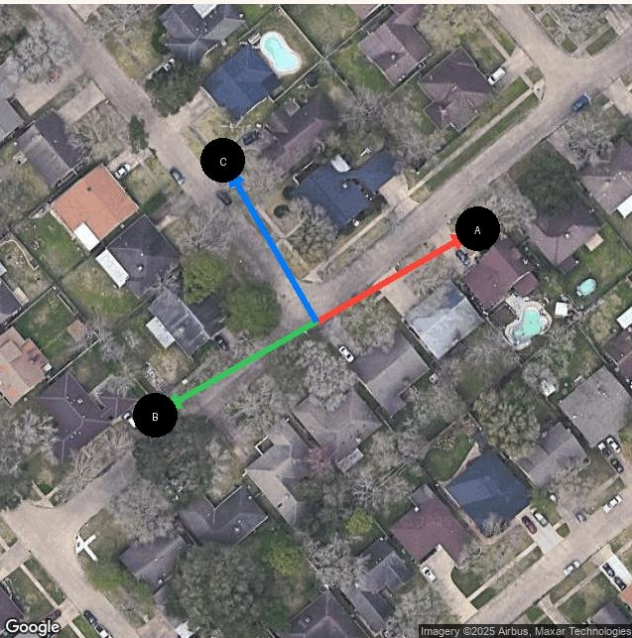


“Both corners appear to have gray roofs; there is no reddish house on the left corner, so B is incorrect.”

Why: Roof color is lighting- and angle-dependent and weakly represented in overhead imagery. A human would discount this cue.

Landmark misbinding

GT: A Pred: B

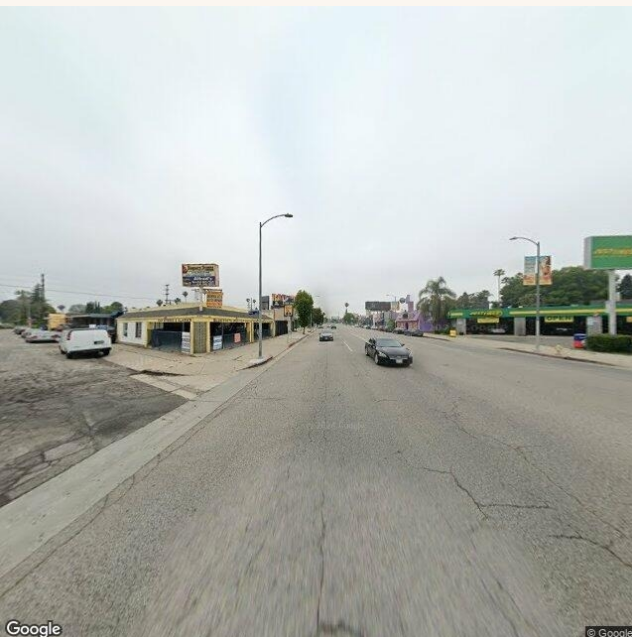
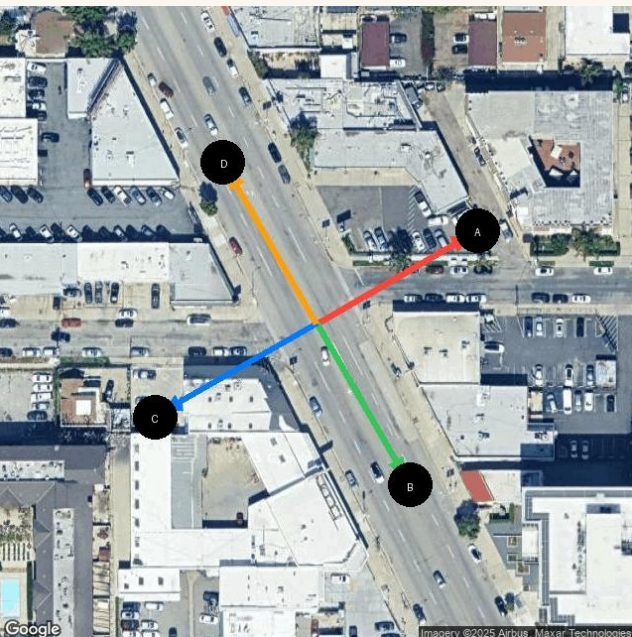


“The corner house has a large swimming pool in its backyard; this does not match the street view, so A is wrong.”

Why: The corner house has *no* pool — one exists two houses down. The model binds a nearby landmark to the wrong spatial anchor.

Symmetry trap

GT: D Pred: B



“Several headings look equally plausible; the model commits on a weak cue instead of resolving the ambiguity.”

Why: At symmetric intersections, models fail to *escalate* to fine cues (lane counts, markings, curb geometry) that would disambiguate.

Takeaways

- m2sv isolates a scalable **map↔street-view alignment** primitive frontier VLMs have *not* solved (best 65% vs human 72/95%).
- Humans beat the best model *and* agree with one another; models lie outside that consensus and collapse on the hardest, symmetric cases.
- SFT+RL on m2sv helps (+10 pts) but transfer is limited — failures are about *adapting reasoning to ambiguity*, not missing data.

Code/data: github.com/YosubShin/m2sv

Try it: m2sv.yosubshin.com